Shao-Ming Ruan

ML ENGINEER · MODEL COMPRESSION & DEPLOYMENT

CONTACT

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SKILLS

LANGUAGES

Python · JavaScript · SQL · C / C++

ML / DEEP LEARNING

PyTorch · Hugging Face · timm · OpenCV · scikit-learn

MODEL COMPRESSION

Pruning (N:M) · Quantization (INT8) · Distillation · ONNX / TensorRT

VISION · LLM · RL

DeepLabV3 · UNet++ · YOLO · RAG / FAISS · Gemma · Llama · PPO / DQN

DATA · WEB · INFRA

pandas · LightGBM · Django · Vue.js · Docker · Git

EDUCATION

National Cheng Kung University 2025 – 2027
M.S., Electrical Engineering — Info. Eng.

National Cheng Kung University 2021 – 2025
B.S., Electrical Engineering

- GPA 4.04 / 4.3 — graduated Top 20%
- Academic Excellence Award (書卷獎)
- Semiconductor Specialization Micro-Program

SELECTED COURSEWORK

- Deep Learning & Applications (A+)
- Computer Vision & Image Processing (A+)
- AI Chip Design & Applications
- System Programming (A+)
- Probability & Statistics (A+)
- Linear Algebra (A+)

PAPER READING · EFFICIENT AI

- MobileNets — depthwise separable conv
- Rethinking the Value of Network Pruning

PROFILE

M.S. student in Electrical Engineering (Information Engineering) at National Cheng Kung University, specializing in efficient AI — making deep-learning models small and fast enough to ship. Hands-on across the full compression stack (channel / token pruning, INT8 quantization, knowledge distillation), with broader work in computer vision, RAG / LLMs, and full-stack Vue / Django builds. Comfortable end-to-end from training and benchmarking to edge deployment, and has led the technical build at hackathons (Best Popularity Award; multiple finalist placements).

PROJECTS

Model Compression Pipeline — Pruning · Quantization · Distillation
2024 – 2025

EFFICIENT-AI RESEARCH · writeup ↗

PyTorch · Network Slimming · PTQ / QAT · timm · ONNX

- Channel-pruned ResNet-50 from 23.5M to 3.65M parameters (−84%) via Network Slimming (BN-γ L1 sparsity); fine-tuning recovered top-1 accuracy to 91.8% (near-lossless).
- Quantized a CNN to INT8 with PTQ and QAT, cutting inference latency from **10.6ms** to **3.5ms** (3.06× faster) while holding 95.38% accuracy.
- Added knowledge distillation (ResNet-50 → ResNet-18, temperature-scaled soft logits) and attention-based ViT token pruning (EViT, keep-rate 0.7) to complete a prune → quantize → distill toolkit.

Edge-AI Elder-Care Companion Robot
2025

SYSTEMS DESIGN

Jetson Nano · TensorRT / TFLite · INT8 · ONNX

- Designed an offline voice-interaction robot for Jetson Nano / Raspberry Pi — Distil-Whisper ASR + a quantized small LLM (LLaMA-3-mini / Phi-3) + YOLOv8-Nano tracking — engineered to ≤1.2s latency, ≤400MB total model size, and ≤15W power.
- Integrated the ASR + LLM + CV modules with INT8 quantization and **2:4** structured pruning, trading off latency, model size, and power at the system level.

PPO vs. DQN for Autonomous Highway Driving
2025

RL RESEARCH · code ↗

PyTorch · PPO · Double DQN · Self-Attention · highway-env

- Reproduced PPO and Double DQN from scratch (GAE, clipped surrogate, replay buffer, target network) on the highway-env benchmark, cross-checked against Stable-Baselines3.
- Ran a parameter-matched Self-Attention vs. MLP ablation (n=5 seeds): episodic returns tied within 5.7% while deployed crash rates differed **~7×** (DQN 11.3% vs. PPO 60–80%), showing episodic return alone is misleading for safety-critical control.

UAV Aerial Semantic Segmentation
2025

KAGGLE COMPETITION · code ↗

PyTorch · DeepLabV3 · UNet++ · ResNet-101

- Built a 16-class UAV segmentation pipeline comparing DeepLabV3 and UNet++ backbones; DeepLabV3-ResNet101 reached a Kaggle public mIoU of 0.58 with a CE + Dice hybrid loss.
- Diagnosed and fixed a normalization / size-alignment bug in the pipeline, recovering the score from mIoU 0.26 to 0.58.

- Knowledge Distillation for Detection (NeurIPS '17)

CERTIFICATIONS

Google Cloud — Generative AI & Cloud Foundations
Vertex AI · LLMs · Prompt Design · 2025

DeepLearning.AI — Deep Learning (Andrew Ng)
Neural Nets · Improving DNNs · Structuring ML · 2023

Machine Learning Foundations — Univ. of Washington
2025

Interactive Programming in Python I–II — Rice
2023

LANGUAGES

English TOEIC 835 (L 455 / R 380)

Mandarin Native

RAG Lecture Q&A System — 2025

LLM / NLP

FAISS · SentenceTransformers · Gemma-3 · Llama-3.2 / Ollama

- Built a retrieval-augmented QA system over lecture decks (PPTX → FAISS vector store); retrieval lifted BLEU 0.028 → 0.110 (**~4**×), ROUGE-L 0.19 → 0.33, and BERTScore 0.848 → 0.888.
- Benchmarked Gemma-3-1B (HF) against Llama-3.2 (Ollama) and showed, via a case study, that RAG can align localized semantics absent from pretraining.

Food-Recognition Health & Finance Platform — 2025

SPARKFUL AI HACKATHON · FINALIST demo ↗

Vue.js · Django · YOLOv11n · REST API

- Finalist among 200+ teams; owned the end-to-end build and retrained YOLOv11n on UECFOOD256 (256 food classes) as the recognition core.
- Shipped a Vue + Django (MVT) app over REST APIs, wiring recognition into health scoring and financial-incentive mechanics.

Public-Health Information Platform — 2024

TAINAN GOV HACKATHON · BEST POPULARITY AWARD

Vue.js · Flask · MySQL

- Best Popularity Award at the 2024 AI Tainan Digital Governance Hackathon (moda / Tainan City); replaced the Health Bureau's error-prone shared-Excel workflow with a role-based web platform.
- Built a Vue + Flask app with RBAC, input validation, timestamped version history, and task tracking for cross-department collaboration.

Suspicious-Account (Fraud) Detection — 2025

AI CUP 2025 · E.SUN

Python · LightGBM · PU Learning · pandas

- Framed positive-only (alert-account) fraud detection as PU Learning (Elkanoto) over a LightGBM base learner; placed 261 / 790.
- Engineered bidirectional (from / to) account-transaction features and handled all-zero columns and sparse labels in a highly imbalanced dataset.

HONORS & COMPETITIONS

- Best Popularity Award — 2024 AI Tainan Digital Governance Hackathon (moda / Tainan City)
- Finalist (200+ teams) — Sparkful “Hack to the Top” AI Hackathon, 2025
- AI CUP 2025 · E.SUN AI Open Competition (suspicious-account detection) — 261 / 790
- Finalist — 2025 Hualien HSH Hackathon
- Academic Excellence Award (書卷獎) — NCKU College of EECS